

The Born–Infeld Saturation Margin of the Chiral Modulus: Antisymmetric Structure, Lorentzian Genus, and Electric Determination of the Generation Split

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Abstract

This note is the companion of the Projective Residue Schur reduction [15] and takes up the first of its open deliverables: the definition of the Lorentzian saturation functional $\mathcal{B}(s)$ along the J_Π -odd modulus that controls the three-generation split coefficient u . We establish three results. First, a structural antisymmetry lemma: the generation modulus is J_Π -odd, and the symmetric square $M = \mathcal{F}\mathcal{F}$ is J_Π -even from birth, so it cannot carry the oriented modulus or the spontaneous $V-A$ branch choice; the primary modulus variable must be an antisymmetric chiral two-form $\mathcal{F}_{\mu\nu}(s)$. Second, we define $\mathcal{B}(s)$ as the Born–Infeld determinantal saturation margin of this two-form, with the overall sign fixed by the admissibility role of $\mathcal{B}$ (projection locking, axiom A4 selects saturated minima), not by matching a Maxwell weak-field expansion; this neutralises the convention sign-trap that would otherwise make the split sign a free choice. Third, the second variation reduces to $\mu_\chi^2 := \partial_s^2 \mathcal{B}(0) = \frac{1}{2} \mathcal{P}_{\mu\nu} \mathcal{P}^{\mu\nu}$, so the existence and stability of the split are governed entirely by the Lorentzian genus of the chiral polarisation $\mathcal{P}$ in the effective metric $g^{\mu\nu} = 2\eta^{\mu\nu}$. The coincidence lock is discharged to a tangential coincidence; the Schur transversality on which it rests is closed in PRS [15] in the present Lorentzian spin stratum, selecting the Schur-transverse branch. Assembling the projector-completion grading, the oriented-cascade lift, and the symbol-compatible spin frame, we then determine the genus: in the Schur-transverse branch the chiral polarisation is electric, so $\mu_\chi^2 < 0$ and the split opens spontaneously. We then extend the radicand to quartic order: the Born–Infeld pseudoscalar cancels on the electric locus, and the amplitude $|u|$ is reduced

to the transverse cubic backreaction of the two-form trajectory, modulo odd reparametrisation, together with the boundary-versus-interior selection of the saturation problem. On the corpus-derived real symplectic cascade the transverse (spin-two) channel of the cubic response vanishes at the metaplectic phase $\gamma = 0$ for every ordering and reference, so the derived amplitude mechanism is Born–Infeld saturation while the split persists, $u \neq 0$; the sixth-order interior lock is doubly conditional, on a non-derived complex phase and on a non-prescribed ordering. Using the companion eliminated-block note, which reduces the split to the normalisation of the single carrier $\partial_s \Delta_\chi(P)|_0$, we then close the boundary-versus-interior selection by admissibility: the saturation contact $\Delta_\chi(s_*) = 0$ is the unique chart-independent A4 lock, the carrier rate being reparametrisation-covariant so that the invariant magnitude is $u(s_*)$, with the interior competitor requiring the non-derived phase and ordering. The magnitude $|u|$ itself is not predicted: it is fixed through the chiral-frontier normalisation $\mathcal{N}_A$, a dictionary-bound quantity.

Keywords: Born–Infeld saturation margin; chiral modulus; antisymmetric two-form; J_Π -equivariance; Schur complement; projection locking; generation split; Lorentzian genus; spontaneous $V - A$ selection.

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1 Position and purpose

The fermionic sub-programme locates the three-generation mass split in the J_3 -odd part of the squared projective endomorphism E_Π^2 restricted to the gauge-singlet generation triplet $\mathbb{C}_{\text{gen}}^3$, parametrised by a single real number u through $E_\Pi^2|_{\mathbb{C}_{\text{gen}}^3} = \text{diag}(1, \frac{1}{2} + u, \frac{1}{2} - u)$, with the even sector $\text{diag}(1, \frac{1}{2}, \frac{1}{2})$ already closed by Born–Infeld parity [8]. The Schur reduction [15] fixed the structural status of u : it is controlled by the $D_\pm$ -transported, generation-projected component of the antiunitary chiral equivariance defect $\Delta_\chi(P) = \pi_{LL} - \tau\pi_{RR}\tau^{-1}$ of the eliminated block $1 - P$, it vanishes for the chirally symmetric minimal non-injectivity, and a non-zero value requires a chiral symmetry-breaking carried by the projection-locking axiom A4. The finite sector was excluded as a source: finite projection locking is J_Π -equivariant, so $u_{\text{fin}} = 0$ for q an odd prime, and the remaining obstruction is entirely Lorentzian [15]. The split is then governed by the chiral curvature $\mu_\chi^2 := \partial_s^2 \mathcal{B}(0)$ of a saturation functional $\mathcal{B}$ along the J_Π -odd modulus s opened by Lorentzian complexification, with the trichotomy: $\mu_\chi^2 > 0$ gives a J_Π -equivariant stable point and $u = 0$; $\mu_\chi^2 < 0$ gives a J_Π -conjugate saturated pair $\pm s_*$ and a spontaneous split $u \neq 0$; $\mu_\chi^2 = 0$ is marginal.

This note settles the structural prerequisites for μ_χ^2 and, assembling published results, determines its sign. Section 2 records that the modulus is J_Π -odd and oriented, and recalls its discrete avatar. Section 3 proves the antisymmetry lemma: the J_Π -odd modulus cannot be represented by the symmetric square $M = \mathcal{F}\mathcal{F}$, and the primary variable is the antisymmetric two-form $\mathcal{F}_{\mu\nu}(s)$. Section 4 defines $\mathcal{B}(s)$ as the Born–Infeld determinantal saturation margin of $\mathcal{F}$, fixes the sign convention by its admissibility role, and isolates the convention trap explicitly. Section 5 reduces the second variation to a Lorentzian norm. Section 6 discharges the coincidence lock to tangential coincidence on the A4 locus and imports the Schur-transversality reduction of [15]. Section 7 determines the genus: assembling the projector-completion grading, the oriented-cascade lift, and the symbol-compatible spin frame, the chiral polarisation is electric in the Schur-transverse branch, so $\mu_\chi^2 < 0$ and $u \neq 0$. Section 8 extends the radicand to quartic order and reduces the amplitude to the sign of the cubic backreaction $\mathcal{P} \cdot \mathcal{R}_\chi$; on the corpus-derived real cascade this transverse channel vanishes path-invariantly, so the derived amplitude mechanism is Born–Infeld saturation and the interior lock is doubly conditional. Section 10 records the epistemic strata.

We work throughout on the flat effective metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ fixed by the Lorentzian metric closure [12], on which the operator-level and spectral-level definitions of u coincide [15].

2 The chiral modulus is J_Π -odd and oriented

The spinorial Born–Infeld parity lift is the antiunitary structure $J_\Pi = \epsilon \circ \text{conj}$ with $\epsilon = i\sigma_y$, giving $J_\Pi^2 = -1$ on the fundamental doublet [9, 15]. On the generation triplet $\mathbb{C}_{\text{gen}}^3 = \text{span}(e_0, e_+, e_-)$ the induced involution $J_\Pi^{(2)}$ satisfies $(J_\Pi^{(2)})^2 = +1$, with $J_\Pi^{(2)}e_0 = -e_0$ and $J_\Pi^{(2)}e_\pm = e_\mp$, and consequently

$$J_\Pi^{(2)} J_3 (J_\Pi^{(2)})^{-1} = -J_3, \quad (1)$$

so that the Born–Infeld conjugation exchanges the two outer generation weights [15].

Lemma 1 (Oddness of the split). *Under $J_\Pi^{(2)}$ the split coefficient transforms as $u \mapsto -u$. If s denotes the J_Π -odd modulus, with $J_\Pi : s \mapsto -s$, then $u = u(s)$ is an odd function, $u(-s) = -u(s)$, and in particular $u(0) = 0$.*

Proof. The triplet endomorphism $E_\Pi^2|_{\mathbb{C}_{\text{gen}}^3} = \text{diag}(1, \frac{1}{2} + u, \frac{1}{2} - u)$ is written in the weight basis (e_0, e_+, e_-) . Since $J_\Pi^{(2)}$ swaps $e_+ \leftrightarrow e_-$ by (1), conjugation by $J_\Pi^{(2)}$ sends this endomorphism to $\text{diag}(1, \frac{1}{2} - u, \frac{1}{2} + u)$, i.e. $u \mapsto -u$. The defect $\Delta_\chi(P)$ and hence u are odd in the modulus s opened by Lorentzian complexification [15], with J_Π acting as $s \mapsto -s$; oddness of $u(s)$ and $u(0) = 0$ follow. $\square$

The orientation carried by the modulus is not an artefact of the chiral frame. Its discrete avatar is the oriented frontier observable of [14]: on the directed outgoing edges $g \rightarrow gs$ of the cascade the central increment is the Heisenberg cocycle $\Delta A_c(g, s) = c(z(s) + a(g)s_b)$, reduced to $ca(g)s_b$ on admissible generators, which measures the dynamic J_3 rather than a free shear. This increment is linear in the orientation datum s_b and therefore odd under orientation reversal. Its symmetric average vanishes identically, $\Theta_{\text{raw}} = 0$ on symmetric shells [10], which is exactly the statement that a symmetrisation annihilates an orientation-odd signal.

3 Antisymmetric structure of the chiral modulus

We now show that the carrier of the J_Π -odd modulus must be an antisymmetric two-form, not a symmetric square. Let $\mathcal{F}_{\mu\nu}(s)$ be a candidate two-form built from the chiral-odd deformation, with the linear expansion

$$\mathcal{F}_{\mu\nu}(s) = s \mathcal{P}_{\mu\nu} + O(s^2), \quad \mathcal{P} := \partial_s \mathcal{F}|_{s=0}, \quad (2)$$

and let

$$M^\mu{}_\nu(s) := \mathcal{F}^{\mu\rho}(s) \mathcal{F}_{\rho\nu}(s) \quad (3)$$

be the symmetric positive combination obtained by squaring it. The antiunitary lift acts on the two-form by $J_\Pi : \mathcal{F}(s) \mapsto \mathcal{F}(-s)$, that is by $\mathcal{F} \mapsto -\mathcal{F}$ at linear order in s , consistently with Lemma 1.

Lemma 2 (Antisymmetric carrier). *The symmetric square $M(s)$ is J_Π -even and stationary at the symmetric point:*

$$M(-s) = M(s), \quad M(0) = 0, \quad \partial_s M(0) = 0. \quad (4)$$

Consequently every scalar functional $\Phi[M]$ is even in s to the order at which M is even and obeys $\partial_s \Phi[M]|_{s=0} = 0$; it assigns the same value to the two J_Π -conjugate branches $\pm s_$ and is therefore blind to the spontaneous $V - A$ sign of u . The split modulus cannot be carried by M . A primary variable that resolves the two branches must change sign under $s \mapsto -s$, hence must be the antisymmetric two-form $\mathcal{F}_{\mu\nu}(s)$.*

Proof. Under J_Π one has $\mathcal{F} \mapsto -\mathcal{F}$ at linear order, so by (3) $M = \mathcal{F}\mathcal{F} \mapsto (-\mathcal{F})(-\mathcal{F}) = M$: the square is J_Π -invariant. Substituting (2) gives $M(s) = s^2 \mathcal{P}^{\mu\rho} \mathcal{P}_{\rho\nu} + O(s^3)$, whence $M(0) = 0$ and $\partial_s M(0) = 0$, which is (4). For any functional $\Phi[M]$ the chain rule gives $\partial_s \Phi[M]|_0 = (\delta\Phi/\delta M) : \partial_s M|_0 = 0$, so Φ has no linear response to the modulus. The spontaneous $V - A$ choice is the sign of $u(s_*) = -u(-s_*)$ by Lemma 1; distinguishing $+s_*$ from $-s_*$ requires a quantity odd in s , which $\Phi[M]$ is not. Therefore M cannot be the primary modulus variable. A two-form $\mathcal{F}_{\mu\nu}(s)$ satisfying $\mathcal{F}(-s) = -\mathcal{F}(s)$ at linear order does change sign and reproduces the orientation-odd increment of the discrete frontier observable [14, 10]; it is the minimal carrier compatible with both the J_Π -oddness of u and the oriented cascade datum. $\square$

Corollary 1 (Role of the symmetric square). *The symmetric combination $M = \mathcal{F}\mathcal{F}$ enters the saturation functional only at second order in s , as a J_Π -even saturating backreaction or higher-order term. It is excluded as the primary modulus variable but admissible as a subleading contribution.*

Proof. By (4) the leading occurrence of M is $O(s^2)$ and J_Π -even, which is the order and parity of a backreaction term rather than of the linear modulus carrier of Lemma 2. $\square$

The lemma resolves the antisymmetric/symmetric fork left open in [15]: the chiral modulus is read off an antisymmetric two-form, and the non-abelian symmetric replacement $M^\mu{}_\nu = \mathcal{F}^{\mu\rho} \mathcal{F}_{\rho\nu}$ used elsewhere to remove

a Pfaffian sign ambiguity is precisely what must be avoided at the level of the primary variable, because removing the sign ambiguity removes the orientation that carries the split.

4 The Born–Infeld saturation margin

We now define the functional whose chiral curvature governs the split. Let $g_{\mu\nu}$ be the effective metric with $g^{\mu\nu} = 2\eta^{\mu\nu}$ [12] and let β be the universal Born–Infeld saturation scale of the bounded-flux completion [2]. Born–Infeld is the unique local completion compatible with a finite transport capacity, and its determinantal action is carried by an antisymmetric field strength [2]; this is the structure into which the antisymmetric two-form $\mathcal{F}_{\mu\nu}(s)$ of Section 3 enters.

Definition 1 (Chiral admissibility radicand and saturation margin). Let

$$\Delta_{\chi}(s) := \frac{-\det(g_{\mu\nu} + \beta^{-1}\mathcal{F}_{\mu\nu}(s))}{-\det g_{\mu\nu}}, \quad (5)$$

the dimensionless ratio in which the metric determinant cancels. Admissibility of the chiral-odd deformation is the reality and non-negativity of the radicand, $\Delta_{\chi}(s) \geq 0$, and saturation is its approach to the bound $\Delta_{\chi}(s) \rightarrow 0$. The Born–Infeld saturation margin along the modulus is

$$\mathcal{B}(s) := \beta^2 \left(\sqrt{\Delta_{\chi}(s)} - 1 \right). \quad (6)$$

The functional $\mathcal{B}$ is fixed in its role, not merely in its formula. By the projection-locking theorem [3] and axiom A4 [1], resolution occurs only where the continuous Born–Infeld saturation meets the discrete shell support; the physically locked chiral configuration is the saturated admissible extremum selected on that support. We adopt the variational reading in which $\mathcal{B}$ is minimised at locking, so that the symmetric point $s = 0$ is the selected configuration exactly when it is a stable minimum of $\mathcal{B}$.

Remark 1 (The convention sign-trap). The determinantal scalar (6) admits an overall sign ambiguity according to the object it names. Read as the Born–Infeld action density, it is extremised by the equations of motion and recovers a Maxwell-type weak-field expansion; read as the saturation margin, it is minimised at the locking event of A4. The two readings differ by an overall sign, and that sign propagates directly into $\partial_s^2 \mathcal{B}(0)$. Choosing the sign so as to obtain a desired trichotomy would make the conclusion an artefact

of convention rather than a derived result. We therefore fix the sign once, by the admissibility role of $\mathcal{B}$ (Definition 1, anchored in [3, 1]), independently of and prior to any knowledge of the sign of the second variation. With this fixing, the sign of the split is never inferred from the convention: it is carried solely by the Lorentzian genus of $\mathcal{P}$ computed in Section 5.

Proposition 1 (Parity of the margin). *$\mathcal{B}(s) = \mathcal{B}(-s)$, so $s = 0$ is automatically a stationary point of $\mathcal{B}$.*

Proof. Under J_Π the two-form obeys $\mathcal{F}(s) \mapsto \mathcal{F}(-s) = -\mathcal{F}(s) + O(s^2)$, and the determinant in (5) depends on $\mathcal{F}$ only through even powers, since for an antisymmetric argument the odd elementary symmetric functions of $g^{-1}\mathcal{F}$ vanish. Hence $\Delta_\chi(s) = \Delta_\chi(-s)$ and $\mathcal{B}(s) = \mathcal{B}(-s)$ by (6). Evenness gives $\partial_s \mathcal{B}(0) = 0$. This is the saturation-functional avatar of the Born–Infeld parity involution [8] and the structural origin of the closed even sector $\text{diag}(1, \frac{1}{2}, \frac{1}{2})$. $\square$

5 Second variation and the Lorentzian reduction

We expand the radicand to the order needed for the chiral curvature. The standard four-dimensional Born–Infeld determinant identity for an antisymmetric two-form gives

$$\Delta_\chi(s) = 1 + \frac{1}{2\beta^2} \mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu} - \frac{1}{16\beta^4} (\mathcal{F}_{\mu\nu} \tilde{\mathcal{F}}^{\mu\nu})^2, \quad (7)$$

where indices are raised with $g^{\mu\nu} = 2\eta^{\mu\nu}$ and $\tilde{\mathcal{F}}$ is the Hodge dual. Substituting the linear expansion (2), the pseudoscalar term is $O(s^4)$ and does not contribute to the second variation, so

$$\Delta_\chi(s) = 1 + \frac{s^2}{2\beta^2} \mathcal{P}_{\mu\nu} \mathcal{P}^{\mu\nu} + O(s^4). \quad (8)$$

Theorem 1 (Lorentzian reduction of the chiral curvature). *With $\mathcal{B}$ as in Definition 1,*

$$\mu_\chi^2 := \partial_s^2 \mathcal{B}(0) = \frac{1}{2} \mathcal{P}_{\mu\nu} \mathcal{P}^{\mu\nu}, \quad (9)$$

so that $\text{sign}(\mu_\chi^2) = \text{sign}(\mathcal{P}_{\mu\nu} \mathcal{P}^{\mu\nu})$, the Lorentzian genus of the chiral polarisation $\mathcal{P}$ in $g^{\mu\nu} = 2\eta^{\mu\nu}$. A spacelike (magnetic-type) $\mathcal{P}$ gives $\mu_\chi^2 > 0$, a stable symmetric point, and $u = 0$; a timelike (electric-type) $\mathcal{P}$ gives $\mu_\chi^2 < 0$, an unstable symmetric point, a J_Π -conjugate saturated pair $\pm s_$, and a spontaneous split $u \neq 0$ whose sign is the $V - A$ choice; a null $\mathcal{P}$ gives $\mu_\chi^2 = 0$, the marginal case.*

Proof. From (6) and (8), $\mathcal{B}(s) = \beta^2 \left(\sqrt{1 + \frac{s^2}{2\beta^2} \mathcal{P}_{\mu\nu} \mathcal{P}^{\mu\nu}} - 1 \right) = \frac{s^2}{4} \mathcal{P}_{\mu\nu} \mathcal{P}^{\mu\nu} + O(s^4)$, giving (9). Because $g^{\mu\nu} = 2\eta^{\mu\nu}$, the contraction $\mathcal{P}_{\mu\nu} \mathcal{P}^{\mu\nu} = 4\eta^{\mu\alpha} \eta^{\nu\beta} \mathcal{P}_{\mu\nu} \mathcal{P}_{\alpha\beta}$ has the same sign as the Minkowski contraction, and the positive overall factor does not affect $\text{sign}(\mu_\chi^2)$. For an antisymmetric two-form the Minkowski contraction is $2(|\mathbf{B}_\mathcal{P}|^2 - |\mathbf{E}_\mathcal{P}|^2)$, positive for a magnetic-type (spacelike-dominated) polarisation and negative for an electric-type (timelike-dominated) one. The mapping to the trichotomy then follows from the variational reading of Definition 1: at $\mu_\chi^2 > 0$ the symmetric point is a stable minimum and is selected ($u = 0$); at $\mu_\chi^2 < 0$ it is a local maximum, and the deformation flows to the saturated admissible pair $\pm s_*$ where, by Lemma 1, $u(\pm s_*) = \mp u(s_*) \neq 0$, the overall sign being the spontaneous $V - A$ selection; at $\mu_\chi^2 = 0$ the second variation is degenerate and the locking character is decided by the leading non-vanishing even coefficient λ_{2k} of $\mathcal{B}(s) = \mathcal{B}(0) + \lambda_{2k} s^{2k} + O(s^{2k+2})$. $\square$

The reduction has a transparent Born–Infeld meaning. A timelike chiral polarisation drives the radicand Δ_χ toward zero, so the chiral-odd deformation reaches saturation at finite $|s|$ and A4 can lock a non-trivial pair; a spacelike polarisation increases the radicand, never saturates, and leaves the symmetric point as the only locked configuration. This is the chiral instance of the canonical Born–Infeld feature that electric-type invariants saturate the bound while magnetic-type invariants do not.

6 Discharge of the coincidence lock

The reduction of Section 5 presupposes that the modulus entering the saturation margin (6) is tied to the J_Π -odd tilt of the eliminated Schur block. A priori these are two distinct quantities: the projector tilt s_{proj} , defined at the level of the projection algebra through the deformation $1 - P(s)$ of the admissible projector $P = \Pi_S^* \Pi_S$, and the saturation modulus s_{BI} , the chiral-odd companion of the Higgs amplitude mode of the saturated condensate. We discharge the lock by transposing the projection-locking theorem to the J_Π -odd sector.

The axiom A4 is an intersection scheme, not a saturation criterion [1, 3]: resolution occurs where the continuous Born–Infeld saturation meets the available discrete support of the projection, and the transition becomes structurally discrete. Transposed to the J_Π -odd sector, a continuous chiral-odd modulus is observable only if it is encoded by the projection. Since the observable space is $\mathcal{O} = \text{Im } \Pi$ and the Schur complement $E_\Pi = -\Pi_S D(1 -$

$P)D\Pi_S^*$ sources the chiral structure through the eliminated block $1 - P$, a chiral-odd Born–Infeld deformation is realisable only as a tilt of the complexified block $1 - P^{\mathbb{C}}$, and conversely a tilt of $1 - P^{\mathbb{C}}$ is dynamically present only on the saturation locus. The two moduli are therefore not independent on the locus. At the finite level this tilt vanishes, $\partial_s(1 - P_{\text{fin}}(s))|_0 = 0$ by the J_{Π} -equivariance of finite locking [15], so the modulus is carried entirely by the Lorentzian completion.

Lemma 3 (Tangential coincidence on the A4 locus). *On the chiral-odd A4 locking locus the Born–Infeld modulus and the Schur tilt are related by a non-degenerate reparametrisation,*

$$s_{\text{BI}} = \phi(s_{\text{proj}}), \quad \phi(0) = 0, \quad \phi'(0) \neq 0, \quad (10)$$

so that A4 identifies their first-order transverse direction up to non-degenerate reparametrisation. This is a tangential identification of the J_{Π} -odd direction, not a strict identification of the parameters.

Corollary 2 (Reparametrisation-invariance of the genus). *The sign of the chiral curvature is invariant under the locking reparametrisation:*

$$\partial_{s_{\text{proj}}}^2 \mathcal{B}(0) = \phi'(0)^2 \partial_{s_{\text{BI}}}^2 \mathcal{B}(0), \quad (11)$$

so $\text{sign}(\mu_{\chi}^2)$ is independent of which modulus parametrises $\mathcal{B}$, and the genus reduction of Theorem 1 is well-posed.

Proof. Equation (11) is the chain rule for $\mathcal{B} \circ \phi$ at $\phi(0) = 0$, using $\partial_s \mathcal{B}(0) = 0$ from Proposition 1; the leading term is $\phi'(0)^2 \partial_{s_{\text{BI}}}^2 \mathcal{B}(0)$, and the positive factor $\phi'(0)^2$ leaves the sign unchanged. $\square$

The first-order projector variation supplies the chiral polarisation explicitly. Differentiating $E_{\Pi}(s) = -\Pi_S D(1 - P(s)) D\Pi_S^*$ at $s = 0$, and using $\partial_s[-(1 - P(s))] = \partial_s P(s)$,

$$\partial_s E_{\Pi}|_0 = \Pi_S D(\partial_s P|_0) D\Pi_S^*, \quad \mathcal{P} = \Pi_S D(\partial_s P|_0) D\Pi_S^*|_{\text{gen}, J_{\Pi}\text{-odd}}. \quad (12)$$

The positive sign is the derivative of $-(1 - P)$, not a hidden conventional choice; it is fixed by the Schur form, in keeping with the convention discipline of Section 4.

Definition 2 (Schur transversality on the J_{Π} -odd eliminated sector). The locking is Schur-transverse if the direction opened by $\partial_s P|_0$ does not lie in the kernel of the Schur transport, that is

$$\Pi_S D(\partial_s P|_0) D\Pi_S^*|_{\text{gen}, J_{\Pi}\text{-odd}} \neq 0, \quad (13)$$

equivalently the non-vanishing of $X \mapsto \Pi_S DX D \Pi_S^*|_{\text{gen}, J_{\Pi}\text{-odd}}$ on the direction $\partial_s P|_0$. Injectivity of this transport on the whole J_{Π} -odd eliminated subspace is sufficient but stronger than required; the lemma depends only on transversality of the opened direction.

The discharge is then a dichotomy.

Proposition 2 (Discharge dichotomy (after the Schur reduction)). *The transverse/null dichotomy is established at operator level in the Schur reduction [15]; transposed to the A_4 locus through Lemma 3 it reads as follows. On the chiral-odd A_4 locus:*

- *if the locking is Schur-transverse, the Born–Infeld modulus and the Schur tilt define the same J_{Π} -odd tangent direction, Lemma 3 holds with $\phi'(0) \neq 0$, and the reduction $\mu_\chi^2 = \frac{1}{2}\mathcal{P}_{\mu\nu}\mathcal{P}^{\mu\nu}$ of Theorem 1 is the genus computation;*
- *if the locking is Schur-null, then $\mathcal{P} = 0$ and $\mu_\chi^2 = 0$ at quadratic order, the marginal branch of the trichotomy of Theorem 1, decided by the first non-vanishing even coefficient λ_{2k} rather than directly by $u = 0$.*

The coincidence lock is thereby discharged to structural status; the Schur transversality it was conditional on is closed in PRS [15], where the projected commutator $[\partial_s P|_0, J_{\Pi}]|_{\text{span}(e_+, e_-)}$ has the exact metaplectic opening $\alpha(t, s) = tsr/\sinh r \not\equiv 0$ with timelike Clifford symbol off the zero-mode locus, selecting the Schur-transverse branch (Proposition 2). By Corollary 2 the eventual genus does not depend on the positive Jacobian $\phi'(0)$.

7 Determination of the genus: electric character of the chiral polarisation

The genus deferred at the end of Section 5 is fixed by the index structure of the chiral polarisation (12) in the completed co-metric, once three published facts are assembled: the spatial/temporal grading of the projector completion [4, 6], the oriented-cascade lift of the J_{Π} -odd modulus [9], and the symbol-compatible spin frame [9].

By the Lichnerowicz form of the projected Dirac square the antisymmetric sector of E_{Π} enters through the Clifford bivector $\frac{1}{2}\gamma^\mu\gamma^\nu\mathcal{F}_{\mu\nu}$ [9], so the polarisation (12) is carried by the Clifford two-form coefficients $\mathcal{P}_{\mu\nu}$, with electric part $\mathbf{E}_{\mathcal{P},i} = \mathcal{P}_{0i}$ in $\gamma^0\gamma^i$ and magnetic part $\mathbf{B}_{\mathcal{P},i} = \frac{1}{2}\epsilon_{ijk}\mathcal{P}_{jk}$ in $\gamma^i\gamma^j$, the decomposition already used in Theorem 1.

Lemma 4 (Index assignment of the central tilt after completion). *In the completed co-metric $g^{\mu\nu} = 2\eta^{\mu\nu}$, with the symbol-compatible spin frame of [9], the J_Π -odd chiral polarisation (12) is purely electric. The dynamic central cascade increment $[X, Y] = Z$ — equivalently the J_3 -odd direction of the oriented cascade — is represented in $\mathcal{P}$ by the temporal-spatial component $\mathcal{P}_{0i}$, while the spatial-spatial component $\mathcal{P}_{ij}$ is the finite static sector and vanishes by finite J_Π -equivariance:*

$$\mathcal{P}_{ij}|_{J_\Pi\text{-odd}} = 0, \quad (\text{Schur-transverse}) \implies \mathcal{P}_{0i}|_{J_\Pi\text{-odd}} \neq 0. \quad (14)$$

Consequently, in the Schur-transverse branch of Proposition 2,

$$\mathcal{P}_{\mu\nu}\mathcal{P}^{\mu\nu} = -2|\mathbf{E}_\mathcal{P}|^2 < 0, \quad \mu_\chi^2 < 0, \quad u \neq 0. \quad (15)$$

Proof. The temporal Clifford generator, as covector then vector. Symbol compatibility [9] fixes the principal symbol of the projected Dirac operator as $\gamma^\mu \xi_\mu$, so the Clifford generators are dual to the cotangent directions: γ^0 is the generator of the temporal covector $d\tau$. The completed co-metric $g^{\mu\nu} = 2\eta^{\mu\nu}$, with $g_{\tau\tau} = -\frac{1}{2} < 0$ [12, 6], raises $d\tau$ to the timelike tangent vector ∂_τ , and [4] identifies ∂_τ with the central Carnot direction Z , which becomes a metric temporal direction only after co-metric completion. Thus $\gamma^0 \leftrightarrow d\tau \xrightarrow{g} \partial_\tau \cong Z$, while the horizontal degree-one generators X, Y are spacelike and orthogonal to Z in the completed co-metric [4], dual to the γ^i .

The magnetic component vanishes. The spatial-spatial Clifford sector $\gamma^i \gamma^j$ pairs two spacelike horizontal directions. The rank-three generation triplet $\mathbb{C}_{\text{gen}}^3$ is the spinorial reading of the same rank-three spatial invariant used in the geometric branch [9]; hence the static weight-preserving generation sector maps to the spatial-spatial Clifford sector. On that sector a static restriction C of E_Π^2 with J_Π -reality and weight-sector preservation $[C, J_3] = 0$ satisfies $C_{++} = C_{--}$ and carries no J_Π -odd modulus [9]; equivalently, the finite tilt $\partial_s(1 - P_{\text{fin}})_{ij}|_0 = 0$ by the J_Π -equivariance of finite locking [15]. Therefore $\mathcal{P}_{ij}|_{J_\Pi\text{-odd}} = 0$ and $\mathbf{B}_\mathcal{P} = 0$.

The electric component is non-zero on the Schur-transverse branch. The single J_Π -odd dynamically active direction is the cascade generator ∂_τ . Its J_Π -odd projection on $\mathbb{C}_{\text{gen}}^3$ is a non-zero multiple of J_3 , the mixing component vanishing in the real symplectic model [9], sourced by the non-commutativity $[X, Y] = Z$; the oriented cascade carries this central degree-two increment as the dynamic J_3 [10, 14]. Since $\partial_\tau \leftrightarrow \gamma^0$, the signal couples γ^0 to the residual spatial structure γ^i and lies in the electric sector $\gamma^0 \gamma^i$. It reaches $\mathcal{P}$ only through the Schur transport of (12): by the Schur transversality (Definition 2, closed in [15]) the opened direction $\partial_s P|_0$ is not in the kernel

of the transport, so the transverse branch of Proposition 2 gives $\mathcal{P} \neq 0$. With $\mathbf{B}_{\mathcal{P}} = 0$ the non-vanishing is carried by the electric component, hence $\mathcal{P}_{0i}|_{J_{\Pi}\text{-odd}} \neq 0$ and $\mathbf{E}_{\mathcal{P}} \neq 0$. In the Schur-null branch the transport annihilates the opened direction and $\mathcal{P} = 0$, the marginal subcase already isolated in Proposition 2.

The genus. With $\mathbf{B}_{\mathcal{P}} = 0$ and, on the Schur-transverse branch, $\mathbf{E}_{\mathcal{P}} \neq 0$, the Minkowski contraction of Theorem 1 is $2(|\mathbf{B}_{\mathcal{P}}|^2 - |\mathbf{E}_{\mathcal{P}}|^2) = -2|\mathbf{E}_{\mathcal{P}}|^2 < 0$, the positive factor from $g^{\mu\nu} = 2\eta^{\mu\nu}$ leaving the sign unchanged. Hence $\mu_{\chi}^2 < 0$ by (9), the symmetric point is unstable, A4 locks a J_{Π} -conjugate pair $\pm s_*$, and $u(\pm s_*) \neq 0$ by Lemma 1. $\square$

Remark 2 (The bracket is an algebraic generation, not a metric index). The Lie bracket $[X, Y] = Z$ is the algebraic generation of the central direction from the horizontal plane, not its metric index. Before completion the symplectic-area reading $X \wedge Y$ is the static finite sector, J_{Π} -protected, with $u_{\text{fin}} = 0$; the completion raises the bracket direction to the temporal axis $\gamma^0 \leftrightarrow d\tau \rightarrow \partial_{\tau} \cong Z$, and the dynamic increment carried by J_3 rides it as the electric component. The same central object changes index-role at completion: magnetic-protected before, electric-active after. The assignment of Lemma 4 is therefore a statement about the completed co-metric, not about the finite Carnot grading.

The genus deferred in Theorem 1 is thereby determined in the Schur-transverse branch: the chiral polarisation is timelike, $\mu_{\chi}^2 < 0$, and the generation split opens spontaneously, $u \neq 0$. What remains is the amplitude $|u|$, the A4 matching of the discrete chiral-frontier normalisation with μ_{χ}^2 , which is not a genus question and is taken up in Section 8.

8 The amplitude frontier

The genus fixes the sign of μ_{χ}^2 and hence the existence of the split; it does not fix the magnitude $|u|$. To locate the saturated pair $\pm s_*$ we extend the radicand beyond the second variation. Because $\mathcal{B}$ is a monotone function of Δ_{χ} through (6), the interior stationary points of $\mathcal{B}$ and of Δ_{χ} coincide, and the square-root concavity changes only the depth of the locked configuration, not its position. The operative object is therefore the quartic of Δ_{χ} , not that of $\mathcal{B}$.

By the parity of Proposition 1 the two-form is an odd function of the modulus, refining the linear expansion (2) to

$$\mathcal{F}_{\mu\nu}(s) = s \mathcal{P}_{\mu\nu} + s^3 \mathcal{R}_{\chi, \mu\nu} + O(s^5), \quad \mathcal{R}_{\chi, \mu\nu} := \frac{1}{3!} \partial_s^3 \mathcal{F}_{\mu\nu} \Big|_{s=0}, \quad (16)$$

where $\mathcal{R}_\chi$ is the cubic response of the chiral two-form, the leading bending of the modulus ray away from its linear direction $\mathcal{P}$. We write $\mathcal{P} \cdot \mathcal{R}_\chi := \mathcal{P}_{\mu\nu} \mathcal{R}_\chi^{\mu\nu}$ for the contraction.

Proposition 3 (Quartic of the chiral radicand). *With the determinantal identity (7) and the odd expansion (16),*

$$\Delta_\chi(s) = 1 + \frac{1}{2\beta^2} \mathcal{P}_{\mu\nu} \mathcal{P}^{\mu\nu} s^2 + \left(\frac{\mathcal{P} \cdot \mathcal{R}_\chi}{\beta^2} - \frac{(\mathcal{P}_{\mu\nu} \tilde{\mathcal{P}}^{\mu\nu})^2}{16\beta^4} \right) s^4 + O(s^6). \quad (17)$$

The quartic carries two contributions of distinct status: a Born–Infeld pseudoscalar term, manifestly non-positive and fixed by $\mathcal{P}$ alone, and a backreaction term $\mathcal{P} \cdot \mathcal{R}_\chi$ whose sign is not fixed by $\mathcal{P}$.

Proof. The exact four-dimensional identity (7) is quadratic in the two invariants $\mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu}$ and $\mathcal{F}_{\mu\nu} \tilde{\mathcal{F}}^{\mu\nu}$. Substituting (16), the scalar invariant is $\mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu} = s^2 \mathcal{P}_{\mu\nu} \mathcal{P}^{\mu\nu} + 2s^4 \mathcal{P} \cdot \mathcal{R}_\chi + O(s^6)$, and the pseudoscalar invariant is $\mathcal{F}_{\mu\nu} \tilde{\mathcal{F}}^{\mu\nu} = s^2 \mathcal{P}_{\mu\nu} \tilde{\mathcal{P}}^{\mu\nu} + O(s^4)$, so its square is $O(s^4)$. Collecting the s^4 coefficients gives (17). The pseudoscalar term $-(\mathcal{P}_{\mu\nu} \tilde{\mathcal{P}}^{\mu\nu})^2/16\beta^4 \leq 0$ depends only on $\mathcal{P}$; the backreaction term is linear in $\mathcal{R}_\chi$, an independent datum of the chiral two-form. $\square$

Born–Infeld therefore supplies no quartic stabiliser of its own: the square root is concave and the pseudoscalar contribution is non-positive. Any interior stabilisation must come from the backreaction $\mathcal{R}_\chi$, and in its absence the modulus runs toward the saturation bound $\Delta_\chi \rightarrow 0$.

On the genus locus the pseudoscalar term vanishes. The index-assignment Lemma 4 gives $\mathbf{B}_\mathcal{P} = 0$, and the pseudoscalar invariant is $\mathcal{P}_{\mu\nu} \tilde{\mathcal{P}}^{\mu\nu} \propto \mathbf{E}_\mathcal{P} \cdot \mathbf{B}_\mathcal{P} = 0$, so the destabilising contribution of Proposition 3 is absent. Writing $\mathcal{P}_{\mu\nu} \mathcal{P}^{\mu\nu} = 2(|\mathbf{B}_\mathcal{P}|^2 - |\mathbf{E}_\mathcal{P}|^2) = -2|\mathbf{E}_\mathcal{P}|^2$ on the electric branch, the radicand reduces to

$$\Delta_\chi(s) = 1 - \frac{|\mathbf{E}_\mathcal{P}|^2}{\beta^2} s^2 + \frac{\mathcal{P} \cdot \mathcal{R}_\chi}{\beta^2} s^4 + O(s^6), \quad (18)$$

so the quartic of the chiral radicand is carried entirely by the backreaction $\mathcal{P} \cdot \mathcal{R}_\chi$.

Proposition 4 (Coordinate-level boundary-versus-interior selection). *Fix an odd local modulus s for the chiral two-form trajectory. On the electric branch the quartic radicand in that parametrisation is governed by the coefficient $\mathcal{P} \cdot \mathcal{R}_\chi$ through the following coordinate-level dichotomy. If $\mathcal{P} \cdot \mathcal{R}_\chi > 0$*

the radicand (18) has an interior minimum candidate at

$$s_*^2 = \frac{|\mathbf{E}_{\mathcal{P}}|^2}{2\mathcal{P} \cdot \mathcal{R}_{\chi}},$$

which defines a physical interior lock only if this minimum survives modulo odd reparametrisation and lies before the first saturation contact, $\Delta_{\chi}(s_*) > 0$. If $\mathcal{P} \cdot \mathcal{R}_{\chi} \leq 0$ no quartic minimum is created in this parametrisation, and the amplitude is fixed by the first contact with the saturation bound, $\Delta_{\chi}(s_*) = 0$.

Proof. Stationarity of (18) at $s > 0$ requires $-2|\mathbf{E}_{\mathcal{P}}|^2\beta^{-2}s + 4(\mathcal{P} \cdot \mathcal{R}_{\chi})\beta^{-2}s^3 = 0$, with the non-trivial root $s_*^2 = |\mathbf{E}_{\mathcal{P}}|^2/(2\mathcal{P} \cdot \mathcal{R}_{\chi})$, real and positive exactly when $\mathcal{P} \cdot \mathcal{R}_{\chi} > 0$. At this root one has

$$\partial_s^2 \Delta_{\chi}(s_*) = -\frac{2|\mathbf{E}_{\mathcal{P}}|^2}{\beta^2} + \frac{12(\mathcal{P} \cdot \mathcal{R}_{\chi})}{\beta^2} s_*^2 = \frac{4|\mathbf{E}_{\mathcal{P}}|^2}{\beta^2} > 0,$$

so the root is an interior minimum of the quartic radicand. For $\mathcal{P} \cdot \mathcal{R}_{\chi} \leq 0$, both the quadratic and quartic coefficients in (18) are non-positive, so no positive interior minimum is created at this order. The admissible region is then terminated by the first contact with the saturation locus $\Delta_{\chi}(s) = 0$, where $\mathcal{B}$ reaches its lower bound $-\beta^2$ and A4 locks. $\square$

This proposition is a statement about the quartic expansion in a fixed odd modulus. Its invariant content is supplied only after quotienting the $\mathcal{P}$ -parallel cubic response by odd reparametrisation, as recorded in the Open paragraph of Section 10.

Corollary 3 (Radial contact in the affine electric gauge). *In the electric Schur-transverse branch, and after quotienting the $\mathcal{P}$ -parallel cubic response by odd reparametrisation, the affine radial contact of the Born-Infeld radicand is fixed by $\Delta_{\chi}(s_*) = 0$. Writing $a := |\mathbf{E}_{\mathcal{P}}|^2/\beta^2$ and $b := \mathcal{P} \cdot \mathcal{R}_{\chi}/\beta^2$, the boundary branch of $\Delta_{\chi}(s) = 1 - as^2 + bs^4 + O(s^6)$ has the positive contact*

$$s_*^2 = \frac{1 - \sqrt{1 - 4b/a^2}}{2b/a} = \frac{1}{a} \left(1 + \frac{b}{a^2} + O(b^2/a^4) \right), \quad (19)$$

whenever the quartic truncation is used away from an interior lock. Equivalently,

$$s_* = \frac{\beta}{|\mathbf{E}_{\mathcal{P}}|} \left(1 + \frac{\mathcal{P} \cdot \mathcal{R}_{\chi} \beta^2}{2|\mathbf{E}_{\mathcal{P}}|^4} + O\left(\frac{(\mathcal{P} \cdot \mathcal{R}_{\chi})^2 \beta^4}{|\mathbf{E}_{\mathcal{P}}|^8} \right) \right). \quad (20)$$

In the affine electric gauge in which the transverse cubic backreaction vanishes, this reduces to the radial contact

$$s_* = \frac{\beta}{|\mathbf{E}_{\mathcal{P}}|}. \quad (21)$$

This fixes the radial factor in the contact split. It does not fix the absolute magnitude of the generation split, because β , $|\mathbf{E}_{\mathcal{P}}|$, and the chiral-frontier normalisation $\mathcal{N}_A$ remain upstream normalisation data.

Proof. The contact condition in the quartic truncation is $1 - as^2 + bs^4 = 0$. With $x = s^2$, the boundary root is the branch continuously connected to $x = 1/a$ as $b \rightarrow 0$,

$$x = \frac{a - \sqrt{a^2 - 4b}}{2b} = \frac{1 - \sqrt{1 - 4b/a^2}}{2b/a}.$$

Expanding the square root gives $x = a^{-1}(1 + b/a^2 + O(b^2/a^4))$, and taking the positive square root gives the stated expression for s_* . When the transverse cubic response vanishes in the affine electric gauge, $b = 0$ in the physical quotient and the first contact is exactly $s_* = \beta/|\mathbf{E}_{\mathcal{P}}|$. $\square$

Remark 3 (Not a Landau closure). The interior branch $\mathcal{P} \cdot \mathcal{R}_{\chi} > 0$ has the local form of a quartic double well, but the stabilisation is not a Landau mechanism: it is conditional on the interior minimum candidate preceding the Born–Infeld saturation wall. The natural locking mechanism remains saturation at $\Delta_{\chi} = 0$, recovered whenever the backreaction fails to turn the radicand before that contact. The amplitude is thus a bounded-flux selection problem, not a polynomial minimisation, and the magnitude $|u|$ follows from s_* only after the discrete chiral-frontier normalisation $\mathcal{N}_A$ maps the modulus to the split through the odd carrier $u(s)$ of Lemma 1.

The cubic response is itself a transported Schur datum. Since the Schur form $E_{\Pi} = -\Pi_S D(1-P)D\Pi_S^*$ is linear in P at fixed D, Π_S [15], the transport acts term by term on the modulus expansion, so

$$\mathcal{R}_{\chi} = \frac{1}{3!} \Pi_S D(\partial_s^3 P|_0) D\Pi_S^*|_{\text{gen}, J_{\Pi}\text{-odd}}, \quad (22)$$

the same transport as the polarisation (12) applied to the third-order tilt. The projector constraint $P^2 = P$ fixes only the P_0 -diagonal part of each derivative in terms of lower ones; the off-diagonal cubic part that sources (22) is unconstrained. Hence the sign of $\mathcal{P} \cdot \mathcal{R}_{\chi}$ is not fixed by $P^2 = P$, nor by the Lorentzian complexification, which supplies the admissible directions but not the curvature of the projector path; it is a genuine open datum.

Hypothesis 1 (Fixed-generator Lorentzian projector orbit). The Lorentzian complexification of the eliminated spinorial block is realised, to the order relevant for the chiral amplitude, by a fixed generator X ,

$$P(s) = e^{sX} P_0 e^{-sX}.$$

Equivalently, the projector path carries no torsion term in the opened J_Π -odd direction up to cubic order.

Remark 4 (Conditional character of the backreaction). Hypothesis 1 is not derived from the present corpus: the complexification fixes the tangent direction X , not the orbit. Under it, $\partial_s^3 P|_0 = \text{ad}_X^2(\partial_s P|_0)$, and on the rank-one J_Π -odd outer-weight direction (the antisymmetric mode $e_+ - e_-$) this collapses to

$$\partial_s^3 P|_0 = -4|b|^2 \partial_s P|_0, \quad \mathcal{R}_\chi = -\frac{2}{3}|b|^2 \mathcal{P},$$

with b the amplitude of X on that mode: the cubic response is parallel to the polarisation. A $\mathcal{P}$ -parallel $\mathcal{R}_\chi$ is pure reparametrisation gauge, removable by the odd modulus transformation $\mathcal{R}_\chi \mapsto \mathcal{R}_\chi + \alpha \mathcal{P}$: the two-form trajectory is then the straight ray $\mathcal{F}(s) = (s - \frac{2}{3}|b|^2 s^3) \mathcal{P}$, the quartic of Δ_χ is gaugeable to the marginal value, and the apparent interior minimum is only the fold of the non-affine modulus. A straight ray has no curvature, so Δ_χ decreases monotonically to the bound and the locked amplitude is the first saturation contact $\Delta_\chi(s_*) = 0$, the boundary branch of Proposition 4, in agreement with Remark 3. A genuine interior lock would require a $\mathcal{P}$ -transverse component of $\mathcal{R}_\chi$, absent on the rank-one mode and not supplied by a fixed-generator orbit; it is part of the open path-curvature datum.

Remark 5 (Path-invariant saturation and the spin obstruction to calibration). Under the adjoint action of the generation $\text{su}(2)$ the endomorphism algebra decomposes into irreducibles, $\text{End}(\mathbb{C}_{\text{gen}}^3) = \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{5}$, with J_3 pure spin one and the weight-two mixing direction that the transverse cubic response $\mathcal{R}_\chi$ must reach pure spin two, in distinct irreducible components. By Schur's lemma an $\text{su}(2)$ -equivariant transport acts on each component by an independent scalar, so calibrating the spin-one (genus) channel of the transport to the electric polarisation does not fix the spin-two (transverse) scalar: the sign of $\mathcal{P} \cdot \mathcal{R}_\chi$ is not reachable by genus calibration, confirming its open status. The transverse cubic response is moreover sourced only by the complex metaplectic phase γ of the Lorentzian complexification, through the unique weight-balanced bracket combination; the real symplectic cascade derived in [9] carries $\gamma = 0$ and its J_3 enters only at second Baker–Campbell–Hausdorff order, so its transverse cubic response vanishes. This vanishing

is path-invariant: for a weight-preserving reference P_0 the phase generator commutes with P_0 , so a phase adjacent to P_0 is a no-op and every ordering of the real cascade gives the same vanishing transverse component, for every reference $P_0 = \text{diag}(p_0, p_1, p_2)$. Hence the saturation branch does not rest on Hypothesis 1: on the corpus-derived real cascade the transverse channel is empty independently of the single-generator model, and the interior lock requires both a complex phase $\gamma \neq 0$, absent from the derived cascade, and a placement of that phase in the cascade, since the transverse cubic datum vanishes when the phase follows the real cascade and is non-zero otherwise, with no placement prescribed.

On the corpus-derived real cascade the boundary-versus-interior selection is therefore resolved in favour of saturation (Remark 5); what remains is the doubly conditional interior lock and the magnitude $|u|$, fixed by the chiral-frontier normalisation $\mathcal{N}_A$, whose dictionary status is recorded in Section 10.

9 Boundary contact as the admissible A4 lock

The companion eliminated-block note [13] reduces the generation split to the normalisation of a single carrier, the transported equivariance defect rate $\partial_s \Delta_\chi(P)|_0$ of the eliminated block, with $\Delta_\chi(P) = \pi_{LL} - \tau \overline{\pi_{RR}} \tau^{-1}$. Throughout this section $\Delta_\chi(P)$ is this equivariance defect (the split carrier), while $\Delta_\chi(s)$ is the admissibility radicand of Definition 1: the two share the chiral subscript but are distinct objects, distinguished by their argument. Proposition 4 stated the boundary-versus-interior dichotomy at coordinate level and deferred its invariant content; we supply that content here and select the boundary by admissibility.

Proposition 5 (The defect rate is reparametrisation-covariant). *Under an odd reparametrisation of the modulus $s \mapsto \phi(s)$, $\phi(0) = 0$, $\phi'(0) \neq 0$, the carrier rate is covariant, not invariant: it rescales by the Jacobian $\phi'(0)$. Hence the rate alone carries no chart-independent magnitude; the invariant datum is the split $u(s_*)$ evaluated at a geometrically-defined modulus point s_* .*

Proof. $\Delta_\chi(P)$ is a function on the modulus line; the chain rule at $s = 0$, using $u(0) = 0$ (Lemma 1), makes the first-order coefficient scale by $\phi'(0)$ between charts, while the value of u at any fixed modulus point is unchanged. This is the first-order companion of the genus covariance of Corollary 2; both are verified by exact computation in [13]. $\square$

Proposition 6 (Chart-independence of the saturation contact). *The first saturation contact $\Delta_\chi(s_*) = 0$ of Proposition 4 is a geometric condition on the radicand: its solution is independent of the chart and of the coefficients of $\mathcal{B}$ beyond the radicand itself. The locked split $u(s_*)$ is therefore reparametrisation-invariant.*

Proof. The contact is the level condition $\Delta_\chi(s_*) = 0$; a level set of a function on the modulus line is a point of the manifold, carried covariantly by any reparametrisation, hence the same physical configuration in every chart. Evaluating the odd carrier u there yields a chart-independent value, verified together with Proposition 5 in [13]. $\square$

Corollary 4 (Boundary contact is the unique admissible lock). *Within the no-circularity discipline — only data derived in the present corpus may enter, with no non-derived complex phase or cascade ordering — the saturation contact $\Delta_\chi(s_*) = 0$ is the unique chart-independent A4 lock. The interior candidate of Proposition 4 is located by the sixth-order coefficient of the radicand, equivalently by the complex metaplectic phase and the cascade ordering left non-derived in the Conditional paragraph of Section 10; a finite truncation of the radicand has, moreover, a chart-dependent extremum. The interior lock is therefore inadmissible unless that phase and ordering are supplied.*

This upgrades the coordinate-level dichotomy of Proposition 4 to an admissibility-level selection: the boundary contact is selected not only on the derived real cascade (path-invariance, Remark 5) but as the unique fence-admissible lock, independent of any chart. The remaining magnitude $|u| = |u(s_*)|$ is thereby reduced, within the fence, to the structural carrier rate of [13] dressed by the contact displacement s_* , with the single honest caveat that a future first-principles derivation of the complex phase and the ordering could reopen an interior competitor.

10 Status of results

Anchored upstream (proved/structural). The Born–Infeld determinantal action carried by an antisymmetric field strength [2]; the even-sector closure and the Born–Infeld parity involution [8]; the finite J_Π -equivariance and $u_{\text{fin}} = 0$ for q an odd prime [15]; the oriented frontier cocycle on directed edges [14] and the symmetric-shell cancellation $\Theta_{\text{raw}} = 0$ [10]; the effective Lorentzian metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ [12]; the projection-locking theorem and axiom A4 [3, 1].

Proved (this note). The chiral modulus is J_Π -odd and the split coefficient $u(s)$ is odd with $u(0) = 0$ (Lemma 1). The symmetric square $M = \mathcal{F}\mathcal{F}$ is J_Π -even and cannot carry the modulus or the spontaneous $V - A$ sign; the primary variable is the antisymmetric two-form $\mathcal{F}_{\mu\nu}(s)$ (Lemma 2, Corollary 1). This resolves the antisymmetric/symmetric fork of [15].

Structural (this note). The definition of $\mathcal{B}(s)$ as the Born–Infeld determinantal saturation margin of $\mathcal{F}$, with the sign fixed by its admissibility role and the convention trap neutralised (Definition 1, Remark 1); the parity $\mathcal{B}(s) = \mathcal{B}(-s)$ (Proposition 1); the reduction $\mu_\chi^2 = \frac{1}{2}\mathcal{P}_{\mu\nu}\mathcal{P}^{\mu\nu}$ and the identification of the trichotomy with the Lorentzian genus of $\mathcal{P}$ (Theorem 1). The discharge of the coincidence lock to structural status: the tangential coincidence of the Born–Infeld modulus and the Schur tilt up to non-degenerate reparametrisation (Lemma 3), the reparametrisation-invariance of the genus (Corollary 2), and the explicit chiral polarisation $\mathcal{P} = \Pi_S D(\partial_s P|_0) D\Pi_S^*|_{\text{gen}, J_\Pi\text{-odd}}$ with its sign fixed by the Schur form (12). The extension of the radicand to quartic order, the non-positivity of its Born–Infeld pseudoscalar contribution, and the cancellation of that contribution on the electric locus (Proposition 3, equation (18)); the reduction of the locked amplitude to the sign of the cubic backreaction $\mathcal{P} \cdot \mathcal{R}_\chi$ through the boundary-versus-interior dichotomy (Proposition 4). The cubic response $\mathcal{R}_\chi$ as the transported third-order Schur tilt $\mathcal{R}_\chi = \frac{1}{3!}\Pi_S D(\partial_s^3 P|_0) D\Pi_S^*$ (equation (22)), and the projector-constraint argument that its sign is a genuine open datum, left free by $P^2 = P$ and by the Lorentzian complexification.

Closed in PRS [15]. Schur transversality on the J_Π -odd eliminated sector (Definition 2) is no longer an A4 residual: it is closed in PRS in the present Lorentzian spin stratum, the projected commutator $[\partial_s P|_0, J_\Pi]|_{\text{span}(e_+, e_-)}$ having the exact metaplectic opening $\alpha(t, s) = tsr/\sinh r \not\equiv 0$ with $\mu(t, s) \equiv 0$ and timelike Clifford symbol off the zero-mode locus, which selects the Schur-transverse branch of Proposition 2. The genus computation of this note is therefore the operative one, and the marginal Schur-null branch does not arise.

Derived (this note, Schur-transverse branch). The Lorentzian genus is electric. The index-assignment lemma fixes $\mathcal{P}_{ij}|_{J_\Pi\text{-odd}} = 0$ and, on the Schur-transverse branch, $\mathcal{P}_{0i}|_{J_\Pi\text{-odd}} \neq 0$ (Lemma 4), assembling the projector-completion grading [4, 6], the oriented-cascade lift [9], and the symbol-compatible spin frame [9]. Hence $\mathcal{P}_{\mu\nu}\mathcal{P}^{\mu\nu} < 0$, $\mu_\chi^2 < 0$, and the generation

split opens spontaneously, $u \neq 0$ (Theorem 1). By Corollary 2 this is well-posed, independent of the positive Jacobian $\phi'(0)$. On the corpus-derived real symplectic cascade the transverse (spin-two) cubic response vanishes path-invariantly at the metaplectic phase $\gamma = 0$ — for every ordering and every reference, by the weight-preserving no-op (Remark 5) — so the derived amplitude mechanism is Born–Infeld saturation: the radicand meets the bound at first contact, with no interior lock, and the split persists, $u \neq 0$. What is settled here is the amplitude mechanism, not the magnitude $|u|$, which remains dictionary-bound (Open).

Conditional (this note). The sixth-order magnetic interior lock — a $\mathcal{P}$ -transverse cubic response, equivalently a non-empty spin-two channel — is doubly conditional on two independent inputs the corpus does not fix. It requires a complex metaplectic phase $\gamma \neq 0$, absent from the derived real cascade; and, even granting the phase, a placement of it in the cascade, since the transverse cubic datum vanishes when the phase follows the real cascade and is non-zero otherwise, with no placement prescribed (Remark 5). The fixed-generator orbit Hypothesis 1 is one special case, on which the cubic tilt collapses on the rank-one J_{Π} -odd outer-weight mode to $\mathcal{R}_{\chi} = -\frac{2}{3}|b|^2\mathcal{P}$, parallel to the polarisation, hence pure reparametrisation gauge (Remark 4). In all cases the interior lock is a candidate, not a computed datum: its specific value is an artefact of a non-prescribed ordering, and the phase produces at most a transverse candidate, never the lock itself. Not concluded.

Selection (this note, Section 9). Within the no-circularity fence the boundary-versus-interior selection is closed by admissibility: the saturation contact $\Delta_{\chi}(s_*) = 0$ is the unique chart-independent A4 lock (Corollary 4), the interior candidate requiring the non-derived complex phase and ordering of the Conditional paragraph. Equivalently, the split carrier $\partial_s \Delta_{\chi}(P)|_0$ of [13] is reparametrisation-covariant, so the invariant magnitude is $u(s_*)$ at the contact. A future first-principles derivation of that phase and ordering could reopen an interior competitor.

Front D factorisation. The eliminated-block computation [13] further factorises the contact split $u_{\partial} = u(s_*)$ into three separated normalisations:

$$u_{\partial} = \sqrt{2} \sigma s_* \frac{T_{J_3}(\partial_s \Delta_{\chi}(P)|_0)}{\langle J_3, J_3 \rangle}.$$

Here $\sqrt{2}\sigma$ is the Born–Infeld even-block factor, from $E_0^2|_{\mathbb{C}_{\text{gen}}^3} = \text{diag}(1, \frac{1}{2}, \frac{1}{2})$, with $\sigma = \pm 1$ the spontaneous $V - A$ branch sign; s_* is the radial A4 contact fixed by $\Delta_\chi(s_*) = 0$, carrying the dependence on the $\mathcal{R}_\chi$ -transverse curvature of the two-form trajectory; and T_{J_3} is the oriented Cartan projection of the $\mathcal{D}^\pm$ -transported chiral defect rate, the angular symplectic-area observable of the oriented-frontier thread — a closed linear functional, not a positive norm — whose normalised evaluation is governed by the dictionary-bound frontier normalisation $\mathcal{N}_A$. The local operator problem is thereby structurally closed: no further selector is hidden in the carrier, in the generator, or in the Born–Infeld even block. The remaining magnitude data are exactly the two upstream quantities already isolated — the radial contact s_* through the A4 radicand curvature, and the angular normalisation of T_{J_3} through $\mathcal{N}_A$ — so the numerical magnitude remains upstream and dictionary-bound.

Radial contact (Front 1). The radial contact factor is now fixed in closed form by Corollary 3: in the affine electric gauge $s_* = \beta/|\mathbf{E}_{\mathcal{P}}|$. The quartic correction is entirely the transverse Schur backreaction $\mathcal{P} \cdot \mathcal{R}_\chi$, whose $\mathcal{P}$ -parallel part is removed by odd reparametrisation (Remark 4). The radial factor of the contact split is thereby settled as a form result, while its numerical magnitude still routes through the upstream normalisations β , $|\mathbf{E}_{\mathcal{P}}|$, and $\mathcal{N}_A$.

Open. The amplitude mechanism is settled (saturation, Derived above); the magnitude $|u|$ is narrowed by Section 8 but not closed. On the derived real cascade the boundary-versus-interior selection is resolved in favour of saturation, so the remaining magnitude datum is the discrete chiral-frontier normalisation $\mathcal{N}_A$ that maps the locked modulus s_* to $|u|$, the cubic response $\mathcal{R}_\chi$ surviving only modulo odd reparametrisation. Equivalently, the open datum is the $\mathcal{P}$ -transverse curvature of the two-form trajectory $\{\mathcal{F}(s)\}$, the part of $\mathcal{R}_\chi$ that fixes the amplitude beyond the genus (informally the torsion of the projector path; Hypothesis 1 sets it to zero), together with the saturation-wall test $\Delta_\chi(s_*) > 0$ and $\mathcal{N}_A$. The cubic response is understood modulo odd reparametrisation of the local modulus: under $s = t + \alpha t^3 + O(t^5)$ one has

$$\mathcal{R}_\chi \mapsto \mathcal{R}_\chi + \alpha \mathcal{P},$$

so the $\mathcal{P}$ -parallel component of $\mathcal{R}_\chi$ is a parametrisation gauge, while the physical datum is the curvature of the two-form trajectory together with $\mathcal{N}_A$ mapping the locked modulus to $|u|$. The residual amplitude problem

is therefore not a parametrisation artefact: the genus is reparametrisation-invariant, and the magnitude is fixed only by the invariant A4 matching of the saturated two-form curve with $\mathcal{N}_A$. This is downstream of the genus and is not a genus question: the genus sign is determined in the Schur-transverse branch (Lemma 4), while the magnitude requires the cubic datum and the normalisation. In the Schur-null branch the quadratic term vanishes and the locking character falls to the marginal branch of Theorem 1, decided by the leading even coefficient λ_{2k} rather than by the genus.

The normalisation $\mathcal{N}_A$ is itself dictionary-bound, not a parameter-free prediction. It is the angular analogue of the radial scale $\kappa = 5/12$ of the exit-deficit dictionary [10, 9], which fixes the split value $\varepsilon = 1/10$ by matching the derived three-level deficit structure to the external ADE ratio 3:2 of a selected case, one of four three-level cases of differing ratios [7]. The first-principles status of $\varepsilon = 1/10$ is therefore exactly that of the ADE case-selection gate, and for the present corpus construction that gate is a no-go: its obstruction is isolated in AOG-Note [11] as an arithmetic orthogonality between Born–Infeld parity and the Galois-spin action, established by a character-table audit of $2I \cong \text{SL}(2, 5)$. The chiral oriented signal that would carry $\mathcal{N}_A$ is well defined — the residual frequency reflection $R_b = W(-I)$ is the central Weil lift, scalar on the chiral carrier, so $[\gamma_5, R_b] = 0$ and the chiral integrand survives the symmetric-frontier cancellation that annihilates the bare cocycle [14, 10] — but evaluating it and comparing to $\varepsilon = 1/10$ would be a dictionary matching, not a test, since $1/10$ is a dictionary value and $\mathcal{N}_A$ its conversion factor. The genuine first-principles front is therefore upstream and structural, not a numerical amplitude: the ADE case selection and the level-to-generation map, the derivation of the projective-resolution growth $\Lambda_{\text{proj}}(n)$ and of the cascade exponent β (only $\beta \leq 1$ established [7, 5]), and the derivation of the transfer constant N_{casc} that fixes $\mathcal{N}_A$ [9].

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$\langle \Delta A_c \rangle_{\partial^+ S_m} \neq 0$, which may fail under residual frontier re-pairing; the discriminating anti-bias control is the symmetrised-frontier cancellation. No normalisation, no dictionary value, no ε enters; the note tests only whether the recursion produces a non-zero, bias-free oriented signal, doi:10.5281/zenodo.20601245.

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